Pressure Regulator Valve – Maintenance Practices

1. Adjustment/Test

A. Bench Pressure Test

Equipment required:

- Pneumatic test rig
- Two pressure gauges 0 to 50 psi
- 1–inch gate valve
- Flowmeter

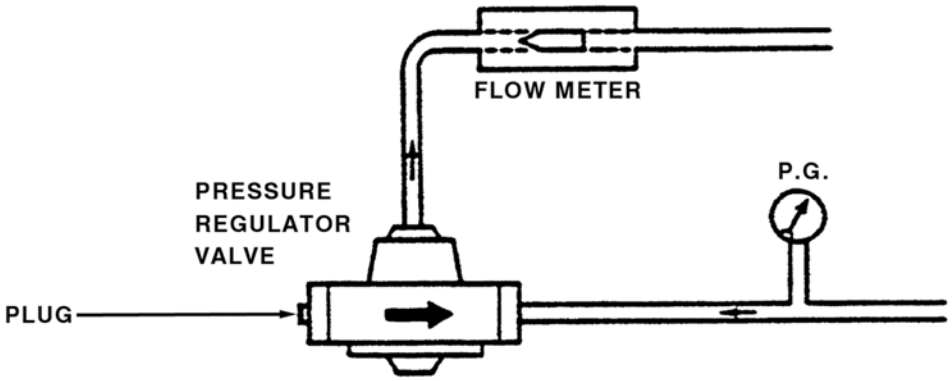
1) Crack and Reseat Pressure

Refer to Figure 1 for details.

a) Install pressure regulator valve as shown in Figure 1.

b) Gradually increase supply pressure to 22 psi; check that relieving flow is 12 cfh minimum.

c) Gradually decrease supply pressure to 18 psi; check that leakage flow is 5 cfh maximum.



Crack and Reseat Pressure – Test Set Up
Figure 1

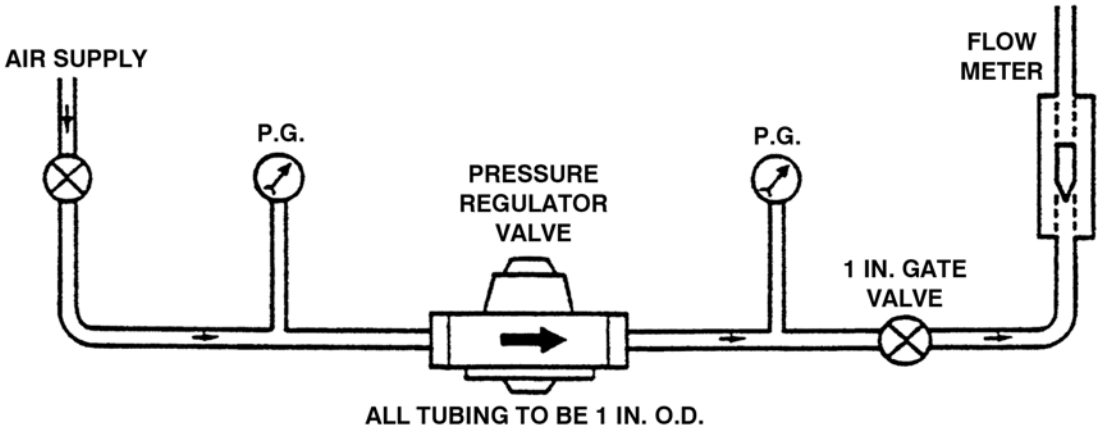
2) Function Test

Refer to Figure 2 for details.

a) Install pressure regulator valve as shown in Figure 2.

b) Apply supply pressure of 30 ± 1 psi and air flow of 5 ± 1/2 cfm: check that regulated air pressure is 18 ± 1 psi.

c) Release air pressure and remove pressure regulator valve. Cap pressure regulator valve connections unless it is to be installed immediately.



Function Test – Test Set Up
Figure 2

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